
Learning Hierarchical Brain Modularity for Alzheimer’s Disease Diagnosis

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Abstract

Functional magnetic resonance imaging (fMRI) is a commonly used technique to measure brain signals. Recent methods that model fMRI data as graphs and leverage graph neural networks (GNNs) to extract features have shown great potential for the early diagnosis of brain disorders such as Alzheimer’s disease (AD). However, existing approaches tend to overlook the modular organization of the brain and enforce rigid constraints with arbitrary predefined cluster sizes. In addition, building inherently explainable GNNs remains challenging. In this paper, we propose the Meta Probabilistic Pooling Graph Neural Network (MPP-GNN) to address these issues. We design a bilevel optimization framework that performs adaptive graph partitioning and utilizes the discovered structural information to guide edge refinement and representation learning. The proposed MPP-GNN is thoroughly validated on the UK Biobank dataset for AD classification and consistently outperforms state-of-the-art baselines. Our analyses at the population, subject, and network levels show that the learned hierarchical brain subgraphs are highly interpretable and match domain knowledge in neuroscience. Our contributions underscore the potential of MPP-GNN for interpretable biomarker discovery in Alzheimer’s disease. The source code is available at <https://github.com/JaxonZ2003/MPP-GNN>.

1 Introduction

The neuroimaging data obtained from fMRI can naturally model brains as graphs. In the standard pipeline, a brain atlas partitions preprocessed fMRI volumes into regions of interest (ROIs) that can be viewed as nodes. The pairwise statistical dependencies between the ROI-level blood-oxygen-level-dependent (BOLD) time series are then estimated to form a functional connectivity (FC) matrix that defines weighted edges [6]. Among various models, graph neural networks (GNNs) are well suited to this representation. The use of GNNs to model brain graphs has not only demonstrated state-of-the-art performances in various brain disorder classification tasks, but also serves as a crucial analytical tool for identifying biomarkers associated with neurological disorders [29].

Despite these advances, GNN-based brain graph analysis faces several challenges: (1) BOLD-derived FC matrices are inherently noisy [27]. (2) The human brain is highly modular, and brain networks in individuals with AD are characterized by disturbed modular structures whose number and composition vary across individuals and disease stages [43]. Yet, most GNN-based community detection methods impose a fixed, predetermined number of clusters that may conflict with the heterogeneous nature of brain organizations [18, 26, 8]. (3) Interpretability in brain graph models remains questionable. Many existing methods rely on post-hoc methods such as saliency maps, or on attention weights, neither of which provides a multi-resolution view aligned with the known hierarchical organization of brain networks [14].

To address these challenges, we propose the meta-Probabilistic Pooling Graph Neural Network (MPP-GNN), an end-to-end framework that jointly refines noisy FC matrices, discovers adaptive community structure, and produces inherent, multi-resolution interpretability. Our main contributions are summarized as follows.

- We formulate brain graph classification as a bilevel optimization problem, where the outer level learns an adaptive community detection strategy to guide the inner-level joint optimization of edge refinement and node representation learning.
- We design an Affinity-Propagation-based Hierarchical Pooling Module (AP-HPM) that discovers multi-resolution community structure without a predetermined cluster count, and a Probabilistic Edge Refinement Module (PERM) that fuses the discovered community structure with node representations through an interpretable logistic gate to produce a differentiable Bernoulli mask.
- Experiments on UK Biobank for AD classification demonstrate that MPP-GNN achieves state-of-the-art performance, and analyses at the population, subject, and network levels show patterns aligned with the established neuroscience literature.

2 Related Work

Graph Structure Learning (GSL) jointly optimizes graph topology and node representations, and has been widely applied to denoise noisy relational data [47]. Existing methods include metric-based approaches using trainable kernels [25, 4, 44], neural approaches like GLCN [15] and NeuralSparse [46], attention-based methods [36, 41], and direct approaches that treat the adjacency matrix as a learnable variable with sparsity [28] or smoothness [17] regularization. In the context of brain graph learning, none explicitly incorporates community structure as an inductive bias for edge refinement.

fMRI Functional Brain Networks. Recent fMRI-specific graph models have increasingly recognized the importance of modular brain organization for disease prediction [35, 33, 32]. BrainNetTF [18] introduces an orthonormal clustering readout, and THC [8] extends it to hierarchical clustering, but their cluster sizes are preset. Community-aware variants such as Com-BrainTF [2] and CAGT [31] rely on predefined community structure. SW-HGL [16] performs Louvain clustering as a separate preprocessing step, preventing community detection from being refined by downstream supervision. Models such as MVS-GCN [37] and IBGNN [7] can suppress noisy connections but do not use community structure as an explicit inductive bias for edge refinement. These limitations motivate a framework that jointly learns adaptive community partitions, community-guided edge denoising, and hierarchical graph representations within a single end-to-end pipeline.

3 Methodology

3.1 Preliminaries

Define an undirected, weighted graph as $G = (\mathbf{X}, \mathbf{A})$, where $\mathbf{X} \in \mathbb{R}^{n \times d}$ is the node-feature matrix with n nodes and d dimensions, and $\mathbf{A} \in \mathbb{R}^{n \times n}$ is the weighted adjacency matrix. Given a graph dataset $\mathcal{D} = \{(G_1, y_1), (G_2, y_2), \dots\}$ where $y_i \in \mathcal{Y}$ is the label corresponding to graph $G_i \in \mathcal{G}$, our goal of graph classification is to learn a model $f_{(\theta, \phi)} : \mathcal{G} \rightarrow \mathcal{Y}$ that maps each graph to a label. We denote $\mathbf{X}^{(0)}$ as the input node features.

Bilevel Optimization. We frame brain graph classification as a bilevel optimization, where the outer level learns an optimal graph-partitioning strategy from Affinity Propagation, and the inner level optimizes hierarchical representations based on the clustering results. Graph partitioning produces discrete cluster assignments that break the gradient flow to downstream inner-level GNN parameters. We adopt the smoothing-based variational optimization (SBO) strategy [23] that bridges the two levels through a distributional parameterization.

Denote θ as the inner-loop parameters, and ϕ as the outer-loop parameters. We learn meta-parameters $\phi := (\boldsymbol{\mu}, \sigma_{meta}^2)$ of a multivariate Gaussian $\mathbf{W}_{meta} \sim \mathcal{N}(\boldsymbol{\mu}, \sigma_{meta}^2 \mathbf{I})$, so that the induced graph-partitioning strategy maximizes the expected score:

$$\max_{\phi} \mathbb{E}_{\mathbf{W}_{meta} \sim \mathcal{N}(\boldsymbol{\mu}, \sigma_{meta}^2 \mathbf{I})} [F(\theta^*(\mathbf{W}_{meta}))], \quad \text{s.t.} \quad \theta^*(\mathbf{W}_{meta}) \in \arg \min_{\theta} \mathcal{L}(\theta; \mathbf{W}_{meta}). \quad (1)$$

At each meta-epoch t , we draw S samples of meta-weights. Each sample induces a distinct AP partition strategy that imposes structural constraints on inner-level learning. The outer-level parameters are updated as:

$$\boldsymbol{\mu}^{(t+1)} = \frac{\sum_s F^{(t,s)} \mathbf{W}_{meta}^{(t,s)}}{\sum_s F^{(t,s)}}, \quad \sigma_{meta}^{(t+1)} = \sqrt{\frac{\sum_s F^{(t,s)} \|\mathbf{W}_{meta}^{(t,s)} - \boldsymbol{\mu}^{(t+1)}\|_2^2}{N_{\mathbf{W}_{meta}} \sum_s F^{(t,s)}}}. \quad (2)$$

The outer-level optimization ends when σ falls below a convergence threshold ϵ or after T meta-epochs. We use AUC as the score F , which is naturally bounded in $[0, 1]$ and appropriate for class-imbalanced clinical settings.

Graph Neural Network. We adopt Graph Convolution Networks (GCNs) [21] as the message propagation operation. For brevity, we omit the meta-optimization indices (t, s) . The node embedding at layer l is computed as:

$$\mathbf{H}^{(l)} = \text{GELU}\left(\text{norm}(\hat{\mathbf{D}}^{(l)-\frac{1}{2}} \hat{\mathbf{A}}^{(l)} \hat{\mathbf{D}}^{(l)-\frac{1}{2}} \mathbf{X}^{(l)} \mathbf{W}^{(l)} + \mathbf{b}^{(l)})\right), \quad (3)$$

where $\hat{\mathbf{A}}^{(l)} = \text{PERM}(\mathbf{A}^{(l)} + \mathbf{I})$ is the refined adjacency matrix from PERM with self-loops, $\hat{\mathbf{D}}_{ii}^{(l)} = \sum_j \hat{\mathbf{A}}_{ij}^{(l)}$, $\mathbf{W}^{(l)}$ is a trainable weight matrix, and $\mathbf{X}^{(l+1)} = \text{AP-HPM}(\mathbf{H}^{(l)})$. norm denotes GraphNorm($\cdot$) [3].

3.2 Affinity Propagation based Hierarchical Pooling

Similarity Matrix Projection. Given a sampled meta-weight matrix $\mathbf{W}_{meta}^{(t,s)}$, we compute meta-features $\mathbf{Z}_{meta}^{(t,s)} = \mathbf{X}^{(0)} \mathbf{W}_{meta}^{(t,s)}$ and construct the initial similarity matrix:

$$\mathbf{S}^{(1)} = \exp(-\mathbf{D}_Z - \lambda_{deg} \mathbf{D}_{deg} - \lambda_{sp} \mathbf{D}_{sp}), \quad (4)$$

where $\mathbf{D}_Z$ is the pairwise ℓ_1 distance in the meta-feature space, $\mathbf{D}_{deg}$ is the absolute degree difference matrix, and $\mathbf{D}_{sp}$ is the shortest-path distance matrix. The topological terms remain fixed while $\mathbf{D}_Z$ evolves across meta-epochs. λ_{deg} and λ_{sp} are hyperparameters.

Hierarchical Pooling of Graphs. Given $\mathbf{S}^{(l)}$, we apply Affinity Propagation (AP) to partition the $n^{(l)}$ nodes into clusters. The AP preference is treated as a hyperparameter and tuned. We denote $\mathbf{Q}^{(l)} \in \{0, 1\}^{n^{(l)} \times n^{(l+1)}}$ as the resulting binary assignment matrix. Because AP does not require a predetermined cluster number, $n^{(l+1)}$ adapts to the structural complexity of each individual graph.

The AP-HPM module coarsens the graph by aggregating node features and inter-cluster connections. Let $\mathbf{D}_q^{(l)} = \text{diag}(\mathbf{Q}^{(l)\top} \mathbf{1})$. The pooled node features are computed by averaging within each cluster: $\mathbf{X}^{(l+1)} = \mathbf{D}_q^{(l)-1} \mathbf{Q}^{(l)\top} \mathbf{H}^{(l)}$. For the pooled adjacency, for two distinct clusters $k_1 \neq k_2$ with at least one connecting edge in the current sparse $\mathbf{E}^{(l)}$:

$$A_{k_1, k_2}^{(l+1)} = \frac{1}{|E_{k_1, k_2}^{(l)}|} \sum_{(i, j) \in E_{k_1, k_2}^{(l)}} A_{ij}^{(l)}. \quad (5)$$

For layers $l > 1$, AP-HPM also pools the similarity matrix as $\mathbf{S}^{(l+1)} = \mathbf{D}_q^{(l)-1} \mathbf{Q}^{(l)\top} \mathbf{S}^{(l)} \mathbf{Q}^{(l)} \mathbf{D}_q^{(l)-1}$. This cascade produces a multi-resolution hierarchy from ROIs to networks to systems. Importantly, the structural components $(\mathbf{S}^{(l)}, \mathbf{Q}^{(l)}, \mathbf{A}^{(l)})$ depend solely on $\mathbf{W}_{meta}$ and can be pre-computed at the onset of each inner-loop optimization.

3.3 Probabilistic Edge Refinement

FC matrices contain false correlations arising from physiological noise and preprocessing artifacts. PERM addresses this by learning an inherently explainable global logistic gate at each hierarchical level l that combines the structural priors obtained from AP-HPM with inner-level node representations.

Global Logistic Edge Gate. We construct three edge-level features for every node pair (i, j) : (i) the similarity metric $S_{ij}^{(l)}$; (ii) a co-membership indicator $C_{ij}^{(l)} \in \{-1, +1\}$ where $C_{ij}^{(l)} = +1$ if i and

j belong to the same cluster; (iii) the cosine similarity between $\mathbf{x}_i^{(l)}$ and $\mathbf{x}_j^{(l)}$. The edge-retention probability is:

$$\mathbf{P}^{(l)} = \sigma(\lambda_1^{(l)} \mathbf{S}^{(l)} + \lambda_2^{(l)} \mathbf{C}^{(l)} + \lambda_3^{(l)} \cos(\mathbf{X}^{(l)}, \mathbf{X}^{(l)\top}) + \lambda_4^{(l)}), \quad (6)$$

where $\lambda_1^{(l)}$ to $\lambda_4^{(l)}$ are learnable scalar parameters. The scalar parameterization keeps the gate highly interpretable: each coefficient directly quantifies the relative contribution of community membership, meta-feature similarity, and feature-level similarity to edge retention.

Stochastic Bernoulli Mask. To convert continuous probabilities $\mathbf{P}^{(l)}$ into a near-discrete mask while preserving gradient flow, we adapt a relaxed Bernoulli sampling scheme [12]. At each forward pass, $\mathbf{U}^{(l)} \sim \mathcal{U}(0, 1)$ element-wise, and the mask is:

$$\mathbf{M}^{(l)} = \frac{1}{2} (\tanh((\mathbf{P}^{(l)} - \mathbf{U}^{(l)})/\tau) + 1). \quad (7)$$

As $\tau \rightarrow 0$, $\mathbf{M}^{(l)}$ converges to a discrete $\{0, 1\}$ Bernoulli mask. The refined adjacency is $\hat{\mathbf{A}}^{(l)} = \mathbf{A}^{(l)} \odot \mathbf{M}^{(l)}$, on which message passing (3) is performed.

3.4 Loss Function

In addition to the classification loss $\mathcal{L}_{\text{cls}}$, we introduce an entropy regularizer $\mathcal{L}_{\text{edge}}$ that encourages near-binary edge decisions:

$$\mathcal{L}_{\text{edge}} = \frac{1}{L} \sum_{l=1}^L \frac{1}{|E^{(l)}|} \sum_{(i,j) \in E^{(l)}} \text{entropy}(\mathbf{P}_{ij}^{(l)}), \quad \mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{edge}} \mathcal{L}_{\text{edge}}, \quad (8)$$

where L is the number of hierarchical levels and λ_{edge} controls regularization strength.

4 Experimental Studies

4.1 Dataset and Preprocessing

We evaluated MPP-GNN on the UK Biobank [34]. We used the Schaefer 2018 atlas to parcellate the brain into 419 ROIs (400 cortices and 19 subcortical regions). We applied PCA to reduce node features $\mathbf{X}^{(0)}$ to 32. For edge weights, we calculated partial correlations and kept the top 30%. UK Biobank is a population epidemiology study of 500,000 participants aged 40 to 69 years; a subset of 100,000 participants is being recruited for multimodal imaging. We used the preprocessed FC matrices following the pipeline in [5]. In datafield 42020, 5,209 participants were identified as AD cases. Among them, only 54 had brain rs-fMRI scans available. Given the substantial class imbalance, we retained all 54 AD subjects and randomly sampled 246 cognitively normal (CN) controls, yielding a final dataset of 300 subjects.

4.2 Experimental Setting

We split data into training, validation, and test sets at a 3:1:1 ratio. At the outer level, we train for 60 meta-epochs with 8 meta-samples per epoch. The inner-level base model has hidden dimensions (64, 32, 16), followed by global mean pooling and a linear classification head. We use Adam [20] with initial learning rate 9×10^{-4} , linearly warmed up for 5 epochs and decayed via cosine annealing. The learning rates for $\lambda_1^{(l)}$ to $\lambda_4^{(l)}$ are 10^{-2} . Batch size is 16, weight decay 5×10^{-3} , and dropout 0.5. All models are trained for up to 300 epochs with early stopping (patience 50). For affinity propagation, max iterations is 3200 and damping factor 0.88; if it fails to converge, we retry with 5000 and 0.95. We fix $\lambda_{\text{deg}} = \lambda_{\text{sp}} = 0.1$. Hyperparameter selection is reported in Appendix C.

4.3 Comparison with Baselines

We compare MPP-GNN with: (1) conventional ML methods: SVM and Random Forest; (2) general-purpose GNN backbones: GCN [21], GAT [36], GIN [40], GraphSAGE [11]; (3) generic graph pooling: DiffPool [42], TopKPool [9], SAGPool [22]; (4) brain-network-specific models: BrainGNN

Table 1: Classification results on the UK Biobank dataset (mean $\pm$ std) over 5-fold-CV. Best in **bold**, second best underlined.

Category	Model	AUC	ACC	Precision	F1
<i>Conventional ML</i>	SVM	74.13 $\pm$ 10.34	67.67 $\pm$ 8.47	80.57 $\pm$ 4.21	70.39 $\pm$ 6.54
	Random Forest	65.00 $\pm$ 9.17	61.00 $\pm$ 14.97	78.16 $\pm$ 3.51	64.07 $\pm$ 13.43
<i>General GNNs</i>	GCN	52.87 $\pm$ 5.57	51.33 $\pm$ 11.99	70.72 $\pm$ 5.20	55.35 $\pm$ 13.48
	GAT	55.58 $\pm$ 8.15	58.00 $\pm$ 9.63	72.16 $\pm$ 4.62	60.94 $\pm$ 7.04
	GIN	56.13 $\pm$ 7.08	60.33 $\pm$ 9.33	72.85 $\pm$ 5.26	62.78 $\pm$ 7.34
	GraphSAGE	55.02 $\pm$ 10.38	67.33 $\pm$ 4.03	72.65 $\pm$ 4.35	69.41 $\pm$ 3.53
	DiffPool	58.77 $\pm$ 10.70	63.67 $\pm$ 10.19	72.25 $\pm$ 3.63	66.06 $\pm$ 7.70
	TopKPool	48.03 $\pm$ 9.99	54.67 $\pm$ 16.24	71.71 $\pm$ 3.67	56.38 $\pm$ 15.47
	SAGPool	44.26 $\pm$ 9.62	57.67 $\pm$ 10.83	67.41 $\pm$ 2.19	60.89 $\pm$ 8.65
<i>Brain networks</i>	BrainNetTransformer	68.70 $\pm$ 10.84	72.00 $\pm$ 9.03	76.48 $\pm$ 4.08	71.02 $\pm$ 4.76
	BrainNetCNN	62.77 $\pm$ 9.84	63.00 $\pm$ 9.45	72.97 $\pm$ 3.24	66.13 $\pm$ 7.67
	BrainGNN	45.18 $\pm$ 4.87	46.33 $\pm$ 17.68	67.51 $\pm$ 2.08	48.47 $\pm$ 13.71
	ContrastPool	63.94 $\pm$ 4.62	70.67 $\pm$ 6.55	73.32 $\pm$ 3.99	70.78 $\pm$ 3.31
	IBGNN	64.57 $\pm$ 5.98	77.00 $\pm$ 6.27	74.67 $\pm$ 5.79	75.22 $\pm$ 5.52
	Contrasformer	59.90 $\pm$ 11.46	78.67 $\pm$ 5.81	74.94 $\pm$ 6.56	<u>75.81</u> $\pm$ 4.57
<i>Ours</i>	MPP-GNN	77.84 $\pm$ 6.24	<u>78.33</u> $\pm$ 4.56	83.28 $\pm$ 3.48	79.48 $\pm$ 3.10

Table 2: Ablation study on UK Biobank. BO = bilevel optimization. Best in **bold**.

AP-HPM	PERM	BO	$\mathcal{L}_{\text{edge}}$	AUC	ACC	$\mathcal{L}_{\text{cls}}$	$\mathcal{L}_{\text{edge}}$	BO	AUC
			✓	54.07 $\pm$ 3.50	66.33 $\pm$ 7.41	✓			62.93 $\pm$ 5.42
✓			✓	60.38 $\pm$ 6.55	68.00 $\pm$ 13.06	✓	✓		70.00 $\pm$ 7.28
✓	✓		✓	70.00 $\pm$ 7.28	74.00 $\pm$ 2.26	✓	✓	✓	77.84 $\pm$ 6.24
✓	✓	✓	✓	77.84 $\pm$ 6.24	78.33 $\pm$ 4.56				

[26], BrainNetCNN [19], Brain Network Transformer (BNT) [18], ContrastPool [38], IBGNN [7], and Contrasformer [39].

The results are summarized in Table 1. MPP-GNN achieves the best AUC, which demonstrates its strong discriminative ability across decision thresholds. This is crucial in the neuroimaging classification task, where class imbalance and threshold sensitivity can make accuracy alone less reliable. MPP-GNN achieves the best AUC, precision, and F1 score. Although Contrasformer has the highest accuracy, its AUC is much lower than that of MPP-GNN, which suggests the high accuracy may not reflect its real ranking quality under class imbalance.

4.4 Ablation Studies

We progressively add the proposed modules onto a plain GCN backbone (Table 2). Disabling all three components reduces the model to a three-layer GCN, with performance close to the GCN baseline. Enabling AP-HPM yields moderate improvement, indicating that community-guided hierarchical pooling provides additional structural information. Adding PERM significantly improves performance, highlighting the severity of noise in raw FC matrices. Finally, enabling bilevel optimization improves AUC to 77.84% and accuracy to 78.33%, demonstrating that outer-level meta-optimization can discover superior community partition strategies. The loss function $\mathcal{L}_{\text{edge}}$ also plays an important role; without it, AUC drops noticeably (Table 2).

4.5 Interpretability Analysis

We further conduct analysis based on the inherently interpretable components of our model from three perspectives: feature importance, population-level discriminative connections, and subject-level hierarchical clusters.

Feature Importance. Tracking the evolution of the logistic gate parameters $\lambda_1^{(l)}$ to $\lambda_4^{(l)}$ across training epochs explains how structural priors and learned representations contribute to edge retention. At the first layer, $\lambda_4^{(1)}$ decreases sharply to a strongly negative value, indicating aggressive pruning of noisy connections; in deeper layers, λ_4 reverses, implying that coarsened graphs contain predominantly

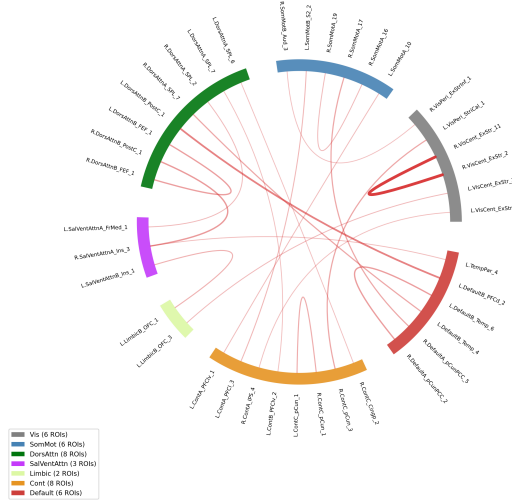


Figure 1: Chord diagram of edges with higher logits in AD than in CN on UK Biobank. Top-20 edges are shown for clarity. Each arc groups ROIs by functional network; each red line represents a top discriminative edge with stronger AD-related signal.

informative embeddings. The cosine-similarity weight λ_3 increases monotonically in deeper layers, while the co-membership weight λ_2 shows the opposite trend—high in the first layer when the input graph is dense and noisy, and near zero after coarsening. The average edge-retention probability converges to 0.02 in the first layer (vast majority of edges pruned) and stabilizes higher in deeper layers. Full visualizations are provided in Appendix A.

Logits and Probability Matrix. Edges with consistently higher logits in one group than the other represent the most informative connections to separate AD from CN. We visualize them in Fig. 1. The most discriminative edges repeatedly involve default mode regions, especially posterior hubs such as the precuneus and posterior cingulate cortex (pCunPCC) and adjacent temporoparietal areas, together with their interactions with control, dorsal attention, somatomotor, and visual systems. This is consistent with the literature on the default mode network in AD and the disruption of posterior default mode network connectivity [10, 30, 1]. The prominent connections among default, control and dorsal attention edges further align with prior evidence that AD disrupts interactions between default mode and attention and executive networks [24]. Somatomotor and visual regions also appear, which may reflect AD-related functional dedifferentiation [13, 45].

Hierarchical Clusters. At the subject level, AP-HPM pools ROIs hierarchically, allowing us to track how an individual region is grouped across layers. Tracing three default-mode ROIs (pCunPCC, medial prefrontal cortex, parahippocampal cortex) through pooling reveals that AD subjects exhibit more compartmentalized aggregation patterns—ROIs remain fragmented across layers—whereas CN subjects show progressively coherent default-related territories. This is consistent with prior literature on default-mode fragmentation in AD [10, 30]. Subject-level visualizations are provided in Appendix B.

5 Conclusion

In this paper, we proposed MPP-GNN. It adopts a bilevel optimization framework to jointly optimize the graph partition strategy and graph representations. MPP-GNN refines noisy connectivity patterns using affinity propagation results as a structural prior. By combining hierarchical community structure with an inherently interpretable design, MPP-GNN provides built-in interpretability without the need for post-hoc explanation. Experiments on UK Biobank show that MPP-GNN consistently outperforms compared baselines and that the patterns discovered are broadly consistent with the neuroscience literature. Limitations include validation on a single dataset and lack of integration with non-imaging modalities. In future work, we will test on additional brain disorder datasets, integrate non-imaging information, and explore other graph partition strategies.

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A Probabilistic Edge-Refinement Dynamics

We provide additional visualizations of PERM’s learned behavior referenced in Section 4.5. Fig. 2 shows the trajectories of the logistic gate coefficients $\lambda_1^{(l)}$ to $\lambda_4^{(l)}$ across training epochs at each pooling layer. Fig. 3 (left) shows the layer-wise mean and standard deviation of the edge-retention probability matrix $P^{(l)}$, and (right) an example subject-level probability matrix at layer 1 where darker entries denote higher retention probabilities.

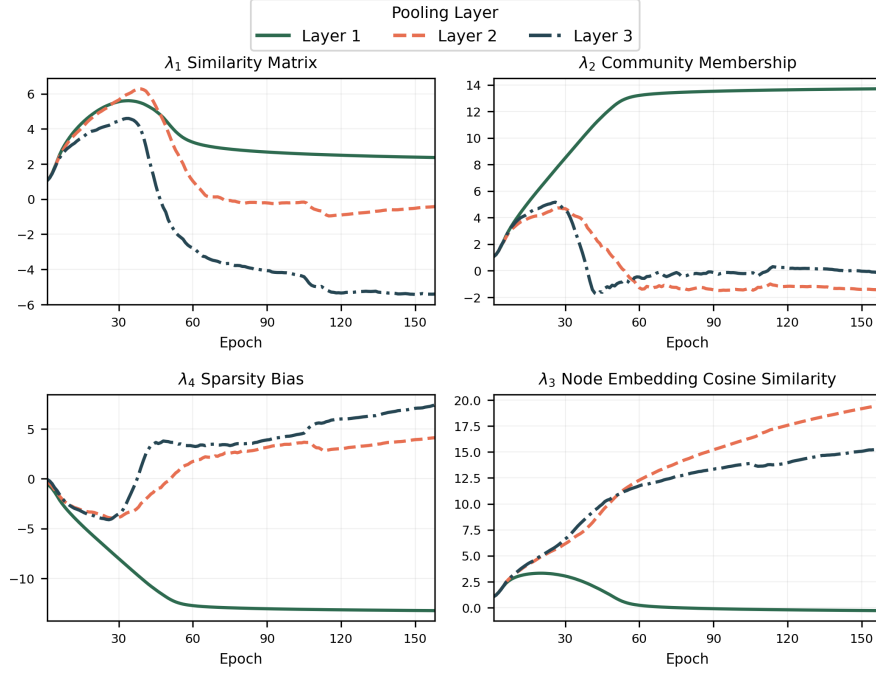


Figure 2: Learned gate coefficients across training epochs. Each line denotes the change of one logistic gate coefficient at the corresponding pooling layer.

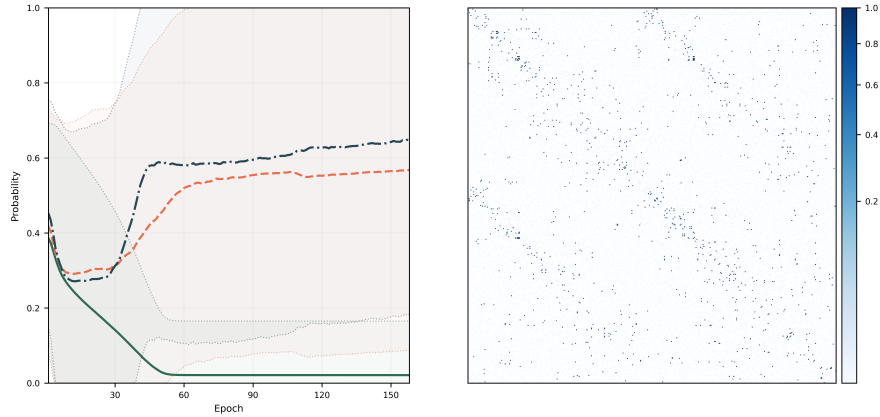


Figure 3: Layer-wise probability statistics (left) and an example subject-level probability matrix at layer 1 (right). The colored band denotes the standard deviation of probability values at the corresponding layer.

B Subject-Level Hierarchical Clustering

Fig. 4 traces three default-mode ROIs (pCunPCC, medial prefrontal cortex PFCm, parahippocampal cortex PHC) through the AP-HPM pooling hierarchy for one AD and one CN subject from the UK Biobank. The ROIs in the AD subject remain fragmented into multiple spatially separated regions across layers, whereas in the CN subject they progressively expand into broader and more spatially coherent default-related territories by Layer 3. This single-subject pattern is consistent with prior literature on default-mode fragmentation and reduced large-scale integration in AD [10, 30] and demonstrates that MPP-GNN can capture clinically relevant disruptions at the individual level.

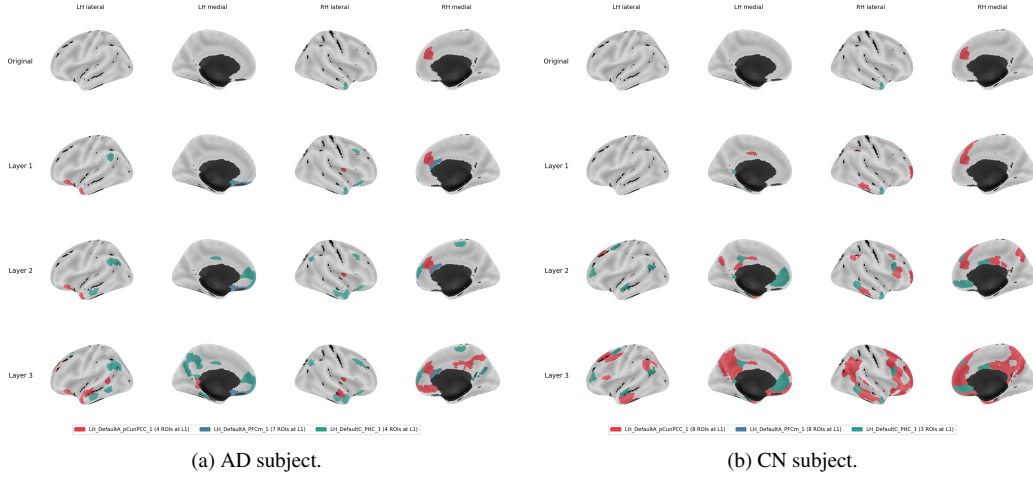


Figure 4: Subject-level tracing of hierarchical community aggregation from three DMN-related communities (pCunPCC, PFCm, PHC) across successive pooling layers on UK Biobank.

C Hyperparameter Analysis

Hyperparameter tuning was conducted on the inner-level base model with the outer-level meta-optimization disabled. We set $\phi = (\mathbf{I}, \mathbf{0})$ and trained with $T = 1$ meta-epoch and $S = 1$ meta-sample. All hyperparameters were tuned using 5-fold cross-validation on a single seed, and we report mean AUC across folds.

AP Preferences. The preference at each AP-HPM layer controls the number of communities discovered. A higher preference yields more, finer-grained communities. We performed a grid search over preference triples and found that performance is sensitive to the granularity of the hierarchical decomposition, with no single layer-wise range guaranteed to yield high performance (Fig. 5). Both per-layer granularity and the cross-layer composition matter for classification. We use (0.30, 0.40, 0.35) for UK Biobank.

Edge-Entropy Loss Weight. The hyperparameter λ_{edge} controls the weight of the edge-entropy regularization. Higher values push retention probabilities away from 0.5, improving interpretability. We swept $\lambda_{\text{edge}} \in [0.1, 1.0]$ and found $\lambda_{\text{edge}} = 0.7$ optimal for UK Biobank.

Bernoulli Temperature. The temperature τ controls the sharpness of the Bernoulli relaxation in PERM. A larger τ pushes the mask toward $\{0, 1\}$, while a smaller τ yields a softer relaxation. We swept $\tau \in [0.1, 1.0]$ and found $\tau = 0.3$ optimal on UK Biobank. AUC drops sharply for $\tau < 0.1$ or $\tau > 1.0$, indicating that moderate sparsification is beneficial but extremes in either direction hurt performance.

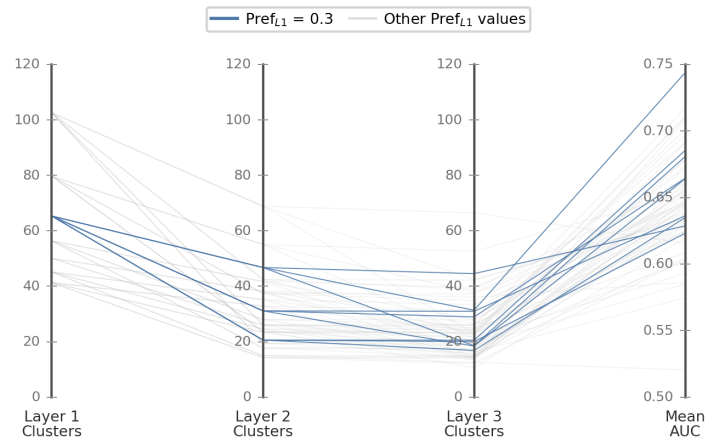


Figure 5: Preference-tuning results on UK Biobank. The first three axes denote the average number of clusters identified by Affinity Propagation at each layer; the rightmost axis is mean AUC.